

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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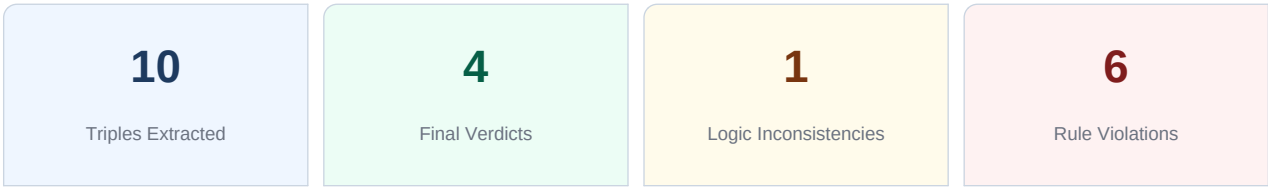
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REPORT OVERVIEW & METHODOLOGY

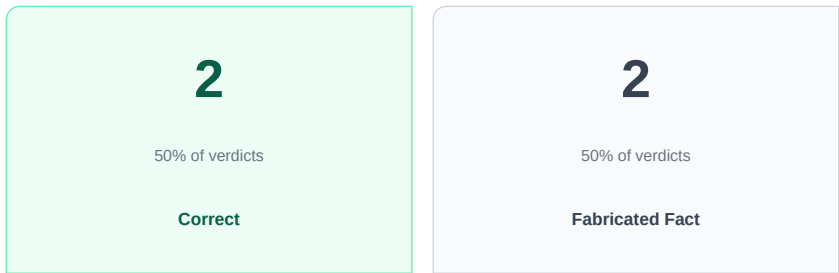
This document outlines the results of the KG-TripleGuard verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	575	HAS_INTEREST_RATE	12.21%	ENTITY_ATTRIBUTED	Denotes the interest rate assigned to the subject entity.
2	580	HAS_INTEREST_RATE	11.91%	ENTITY_ATTRIBUTED	Denotes the interest rate assigned to the subject entity.
3	579	HAS_INTEREST_RATE	12.03%	ENTITY_ATTRIBUTED	Denotes the interest rate assigned to the subject entity.
4	579	HAS_INTEREST_RATE	11.91%	ENTITY_ATTRIBUTED	Denotes the interest rate assigned to the subject entity.
5	890	HAS_INTEREST_RATE	4.36%	ENTITY_ATTRIBUTED	Denotes the interest rate assigned to the subject entity.
6	890	HAS_RISK_TIER	super prime	ENTITY_ATTRIBUTED	Denotes the risk tier classification of the subject entity.
7	890	HAS_RISK_TIER	above super prime	ENTITY_ATTRIBUTED	Denotes the risk tier classification of the subject entity, suggesting it is better than super prime.
8	890	HAS_DOWN_PAYMENT	2.0%	ENTITY_ATTRIBUTED	Specifies the down payment percentage required for the subject entity.
9	290	HAS_LOAN_TERM	12	ENTITY_ATTRIBUTED	Denotes the duration or term of the loan for the subject entity.
10	850	HAS_DOWN_PAYMENT	53%	ENTITY_ATTRIBUTED	Specifies the down payment percentage required for the subject entity.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	12.03%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
890	HAS_INTEREST_RATE	4.36%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
890	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_RISK_TIER	above super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_DOWN_PAYMENT	2.0%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
290	HAS_LOAN_TERM	12	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	HIGH
850	HAS_DOWN_PAYMENT	53%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4.36%	c_interest_rate_range	Subject (credit score) 890 > 850, which violates the constraint that credit score must be between 300 and 850.
factually_inconsistent	RULE_VIOLATION	890	RISK_TIER_CLASSIFIED_AS_BORROWER_RANGE_LEVEL	super prime	c_risk_outside_range	Subject (credit score) 890 > 850, which violates the constraint that credit score must be between 300 and 850.
factually_inconsistent	RULE_VIOLATION	890	RISK_TIER_CLASSIFIED_AS_BORROWER_RANGE_LEVEL	above super prime (qualifier: above)	c_risk_outside_range	Subject (credit score) 890 > 850, which violates the constraint that credit score must be between 300 and 850.
factually_inconsistent	RULE_VIOLATION	890	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	2.0%	c_down_payment_range	Subject (credit score) 890 > 850, which violates the constraint that credit score must be between 300 and 850.
factually_inconsistent	RULE_VIOLATION	290	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12	c_loan_term_enum	Subject (credit score) 290 < 300, which violates the constraint that credit score must be between 300 and 850.
factually_inconsistent	RULE_VIOLATION	850	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	53%	c_down_payment_range	Object (down payment %) 53 > 50, which violates the constraint that down payment must be between 0 and 50.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
575	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	Correct	factually_consistent	—	SEMANANTIC	—	●●●●●●●● ■ 0.80	The triple matches a DB candidate with a minor semantic difference in the object. Ev: {"subject": "575", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "12.21"}

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	Correct	factually_consistent	—	SEMANANTIC	—	●●●●●●●● ●■ 0.80	The triple matches a DB candidate with a minor semantic difference in the object. Ev: {"subject": "580", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "11.91"}
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	—	■■■■■■■■ ■■■ 0.00	The triple does not match any DB candidate and the predicate is known in the schema. Ev: []
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	—	■■■■■■■■ ■■■ 0.00	The triple does not match any DB candidate and the predicate is known in the schema. Ev: []

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

Label	Type	Severity	Explanation	Triples Involved
logically inconsistent	DIRECT_PROPOSITIONAL_CONTRADICTION	HIGH	The same subject '579' has two different interest rates asserted for the 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE' predicate, which is a functional property.	579 -> INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE -> 12.03% [logically_inconsistent] 579 -> INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE -> 11.91% [logically_inconsistent]

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.